

**In one line: Accessibility:** VoiceOver walks a tree of *elements* and speaks each one's **label, value, traits, hint**; text must scale with **Dynamic Type**; honour user settings (Reduce Motion, contrast). **Localization:** every user-visible string is a *whole sentence* in a **String Catalog**, arguments are **positional**, plurals follow **CLDR** rules, layout is **leading/trailing**, and numbers/dates go through **formatters**.

## Accessibility

- **Label** = what it is ("Volume" — never "Volume button", the trait says button). **Value** = current state ("70%"). **Traits** = role/state (.button .header .selected .adjustable .notEnabled). **Hint** = result of acting, spoken last, after a pause; users can turn hints off.
- `isAccessibilityElement` = true makes a view a leaf; on a container it *hides the children*. Order children with `accessibilityElements` = [ ... ]. Decorative: `.accessibilityHidden(true)`.
- **Grouping** (SwiftUI) `.accessibilityElement(children:)`:  
 .combine (one swipe, merged) · .ignore (you label it) ·  
 .contain (keep children, group them).
- .adjustable needs `accessibilityIncrement()` / `Decrement()` (SwiftUI `.accessibilityAdjustableAction`).
- **Custom actions** (`UIAccessibilityCustomAction`, `.accessibilityAction(named:)`) expose swipe/long-press actions via the **Actions rotor**; custom **rotors** (`.accessibilityRotor`) jump between headings/links. Announce: `UIAccessibility.post(notification: .announcement, ...)`.
- `accessibilityIdentifier` is for **UI tests**: stable, not localized, never spoken.
- **Dynamic Type**: `UIFont.preferredFont(forTextStyle:)` or `UIFontMetrics(forTextStyle:).scaledFont(for:)`, plus `adjustsFontForContentSizeCategory` = true (default false) and `numberOfLines` = 0. SwiftUI text styles scale by themselves; custom sizes: `@ScaledMetric`. At the accessibility sizes, stack vertically.
- **Settings**: Reduce Motion (`\.accessibilityReduceMotion`: swap zooms for fades), Reduce Transparency, Increase Contrast, Bold Text. Never colour alone.
- **Audit**: Accessibility Inspector · `XCUUIApplication().performAccessibilityAudit()` (Xcode 15).

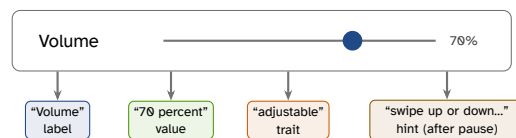
## Localization

- **String Catalog** .xcstrings (Xcode 15, JSON): keys extracted at build, per-language state, plural + device variants inline. Replaces `.strings` + `.stringsdict`.
- Code: `String(localized: "key", defaultValue:, comment:)` (iOS 15), `LocalizedStringResource`; SwiftUI `Text("literal")` is a `LocalizedStringKey`. Packages: `bundle: .module`. Always a `comment:`.
- **Plurals**: "Vary by plural" — the language's CLDR categories (table). Never count == 1 ? :.
- **RTL**: leading/trailing + .natural alignment mirror; left/right never do. Opt out per view with `semanticContentAttribute(.playback for media controls)`; SwiftUI `\.layoutDirection`.
- **Formatters**: `date.formatted(...)`, `.currency(code:)`, `Measurement(km↔mi)`. Formats follow `Locale.current` (region settings); the UI language is `Bundle.main.preferredLocalizations`.
- **Test**: pseudolanguages (Accented, Bounded, Double-Length, Right-to-Left); vendors get *Export Localizations* (.xcloc = XLIFF + context).

## A volume row, done right

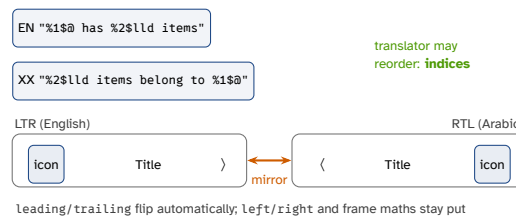
```
HStack {
    // a volume row
    Image(systemName: "speaker.wave.2").accessibilityHidden(true)
    Text("Volume") // LocalizedStringKey
    Slider(value: $level)
}
.accessibilityElement(children: .ignore)
.accessibilityLabel("Volume") // no "slider" - trait adds it
.accessibilityValue(Text(level, format: .percent))
.accessibilityAdjustableAction { // swipe up / down
    level += $0 == .increment ? 0.1 : -0.1
}
.accessibilityIdentifier("settings.volume") // tests only
// catalog key "%lld files deleted", varied by plural
let msg = String(localized: "\{(count) files deleted",
    comment: "Toast after bulk delete")
```

## What VoiceOver says



identifier "settings.volume" — never spoken, XCUITest only

## Positional arguments & mirroring



## Numbers

|                                 |                                                                                                                                                                                                     |                                                         |    |       |    |           |         |                    |    |                             |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|----|-------|----|-----------|---------|--------------------|----|-----------------------------|
| <b>Contrast</b> (WCAG AA)       | text ≤ 17 pt<br>≥ 18 pt, or bold                                                                                                                                                                    | <b>4.5:1</b><br><b>3:1</b>                              |    |       |    |           |         |                    |    |                             |
| <b>Hit target</b> default / min | iOS, iPadOS, watchOS<br>visionOS<br>macOS · tvOS                                                                                                                                                    | <b>44</b> / 28 pt<br>60 / 28 pt<br>28 / 20 · 66 / 56 pt |    |       |    |           |         |                    |    |                             |
| <b>CLDR plurals</b>             | <table><tr><td>ja</td><td>other</td></tr><tr><td>en</td><td>one other</td></tr><tr><td>pl · ru</td><td>one few many other</td></tr><tr><td>ar</td><td>zero one two few many other</td></tr></table> |                                                         | ja | other | en | one other | pl · ru | one few many other | ar | zero one two few many other |
| ja                              | other                                                                                                                                                                                               |                                                         |    |       |    |           |         |                    |    |                             |
| en                              | one other                                                                                                                                                                                           |                                                         |    |       |    |           |         |                    |    |                             |
| pl · ru                         | one few many other                                                                                                                                                                                  |                                                         |    |       |    |           |         |                    |    |                             |
| ar                              | zero one two few many other                                                                                                                                                                         |                                                         |    |       |    |           |         |                    |    |                             |

## Traps

- ! "Play button" as label → "Play button, button".
- ! `isAccessibilityElement` on a container hides its children.
- ! `preferredFont` without the `adjustsFont...` flag → never re-scales.
- ! UI tests matching localized labels — use `accessibilityIdentifier`.
- ! `"\{(n) " + "items"` — breaks word order *and* plurals.
- ! `Text(aStringVar)` is *not* localized — only literals / keys.

## Remember

**"L-V-T-H"** — Label, Value, Trait, Hint (the order VoiceOver speaks). **Localize sentences, not words; lead, don't left.**